MATHEMATICAL OPTIMIZATION: LAB

Abstract. Tasks for the LAB

1. LAB 1

1. Install Julia + Jupyter Lab + JuMP solver framework + SCS solver and work through the linear programming Jump tutorial https://jump.dev/JuMP.jl/stable/tutorials/getting_started/getting_started_with_JuMP/

   Apart from Julia itself (and perhaps jupyter lab) you should be able to install most of this through the Julia package manager.

2. LAB 2

2. Check *strong duality* and *complementary slackness* for numerical solutions.
3. Dualize an infeasible LP and obtain infeasibility certificate

3. LAB 3

4. Impement LP relaxations for maximum perfect matching and vertex cover.

4. LAB 4

5. Implement the relaxations FSTAB, QSTAB for the stable set polytope.
6. Let $P = \text{conv}\{(0,0),(2,0),(0,1),(2/3,2/3)\}$. Describe the Anti-blocker P^*.
7. Recall:

   Theorem 1 (Chvatal for positive orthant: also rounds down the lhs)**.** *Given*

$$\min\{c^T x \,:\, Ax \leq b, x \in \mathbb{Z}^n_+\} \tag{1}$$

   where $A \in Z^{m\times n}, b \in \mathbb{Z}^m, c \in \mathbb{R}$*. Then the inequality*

$$\lfloor \lambda^T A \rfloor x \leq \lfloor \lambda^T b \rfloor \tag{2}$$

   is valid for P_I *for all* $\lambda \in \mathbb{R}^m_{\geq 0}$.

   Make numerical iterative Gomory-Chvatal cuts for

$$\begin{aligned} \max \quad & 2x_1 + 5x_2 \qquad (3)\\ \text{subject to} \quad & 4x_1 + x_2 \leq 28\\ & x_1 + 4x_2 \leq 27\\ & x_1 - x_2 \leq 1\\ & x_1, x_2 \geq 0 \qquad (4) \end{aligned}$$

   Find the numerical Gomory-Chvatal Closure P'!

   (Example is from: https://coral.ise.lehigh.edu/ ted/files/ie418/lectures/Lecture13.pdf. Optimal basic feasible solution is: $x_1^* = 17/3, x_2^* = 16/3$ with $\alpha^* = 38$. Optimal IP solution is $\alpha^*_{IP} = 37.5$.

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